

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

6th Semester of B. Pharm Examination

University Theory Examination April-May 2016

PHARMACEUTICAL BIOCHEMISTRY-II (PH311)

Date: 03.05.2016, Tuesday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any **TWO** of the followings

- | | | |
|----------|---|----|
| A | 1. Give <i>de novo</i> pathway for the biosynthesis of pyrimidine nucleotides. | 07 |
| | 2. Discuss the types of chemical bonds present in the nucleic acids and its role in the formation of DNA and RNA. | 03 |
| B | 1. Write a note on degradation of purine nucleotides. | 04 |
| | 2. What do you mean by porphyrins? Give biosynthesis of porphyrins. | 06 |
| C | 1. Discuss metabolic disorders related to porphyrins. | 04 |
| | 2. Give an account on sulphur cycle. | 06 |

Q 2 Attempt any **TWO** of the followings

- | | | |
|----------|---|----|
| A | What do you mean by DNA replication process? Enlist the name and function of enzymes involved in DNA replication process. Give an account on significance of DNA replication. | 10 |
| B | Write a short note on following: | 10 |
| | 1. Genetic code | |
| | 2. DNA repair mechanisms | |
| C | 1. Discuss N ₂ Cycle with suitable reactions. | 06 |
| | 2. Explain degradation of pathway for heme. | 04 |

SECTION- II

Q 3 Attempt any **TWO** of the following

- A** Classify vitamins with structures and add a detailed note on role of vitamine A in vision. 10
- B** What does the chemi-osmotic theory of oxidative phosphorylation states? 10
Explain the main proton circuit for oxidative phosphorylation and structure of ATP Synthase enzyme.
- C** 1. Name various components of respiratory chain and explain transfer of reducing equivalents through coenzyme Q. 04
2. Classify kidney function tests and explain any two clearance tests. 06

Q 4 Attempt any **TWO** of the following

- A** Write a short note on following: 10
1. RIA
2. ELISA
- B** What do you mean by electrophoresis? Which type of molecules can be subjected to the technique and why? Explain in detail various types of paper electrophoresis. 10
- C** Add a short note on following: 10
1. Uncouplers of oxidative phosphorylation
2. PCR